Experiment 8: Write python program to Depth-First Search

Aim:

To write a Python program to implement the Depth-First Search (DFS) algorithm.

Algorithm:

1. Start from the initial node.
2. Visit and print the node.
3. Mark the node as visited.
4. Visit an unvisited adjacent node.
5. Repeat until all reachable nodes are visited.

Program:

```
graph = {
   'A': ['B', 'C'],
   'B': ['D', 'E'],
   'C': ['F'],
   'D': [],
   'E': [],
   'F': []
}

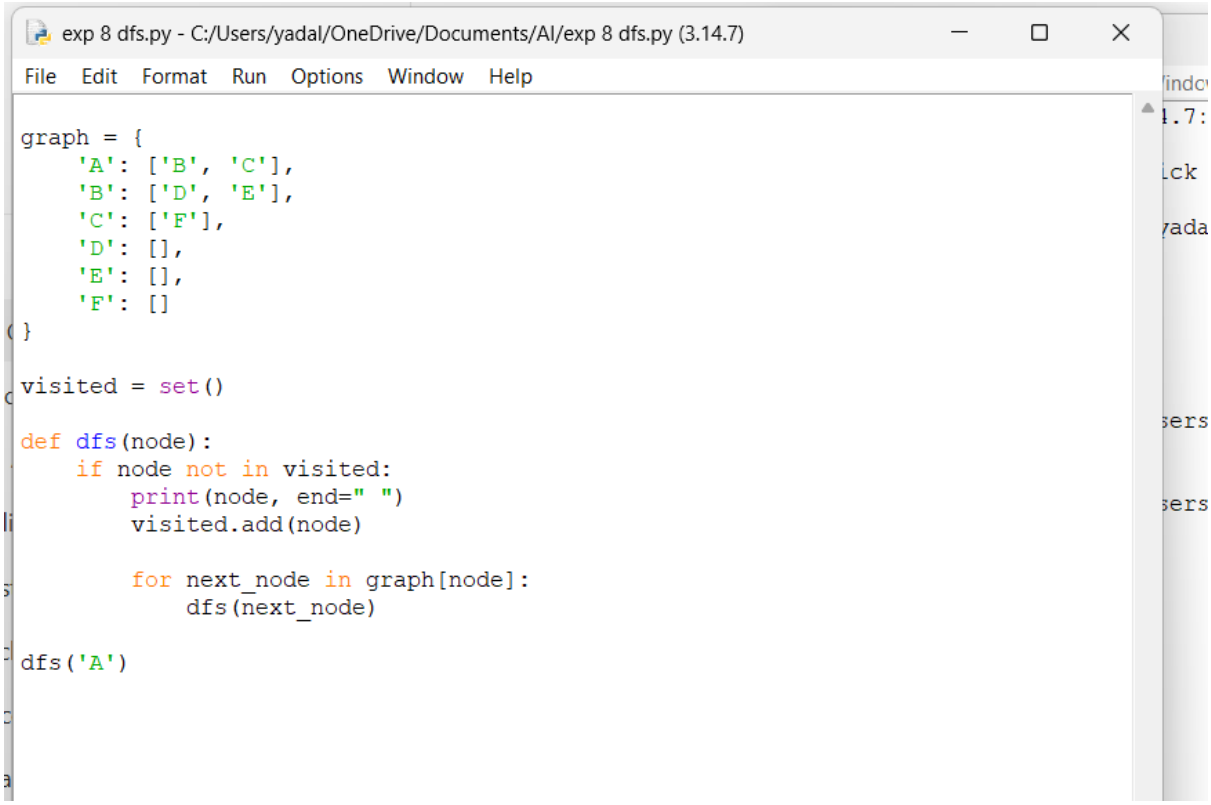
visited = set()

def dfs(node):
   if node not in visited:
      print(node, end=" ")
      visited.add(node)

      for next_node in graph[node]:
```

dfs(next_node)

dfs('A')



Output:

```
========== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 8 dfs.py =========
A B D E C F
>>>
```

Result:

Thus, the Python program to implement Depth-First Search (DFS) was executed successfully.